RANK-EIGHT GROTHENDIECK RINGS THROUGH MULTIPLICITY TWO: CRITERIA, MODELS, AND THE UNRESOLVED FRONTIER

Abstract. We analyze general complex categorification of every rank-eight fusion ring of multiplicity at most two in a complete census. An ordered collection of exact tensor and arithmetic tests excludes 846 of the 973 rings. Full multiplication-tensor comparisons identify unitary categories for 72 more. The remaining 55 rings are recorded as unresolved, including 3 already at multiplicity one; thus this is an explicit categorification frontier, not a complete classification. A rank-eight Wedderburn lemma extends the formal-codegree d-number obstruction to noncommutative rings. Every decision and every positive tensor comparison accompanies the manuscript in a checked machine-readable ledger.

1. Introduction and exact scope

The fusion-ring census [11] supplies all 96 rank-eight based fusion rings of exact multiplicity one and all 877 of exact multiplicity two. A fusion ring need not be the Grothendieck ring of a fusion category. We work over $\mathbb{C}$ and do not assume pivotality, sphericality, braiding, commutativity or unitarity when generally excluding a ring. A positive construction is a unitary category with the exact based-isomorphism type of its full multiplication tensor.

The lower-rank criteria and their finite verifiers are developed in [12, 13, 14, 15, 16, 17]. We cite their theorems and do not reprove them. These new manuscripts are cited by title, with no byline. Our rank-eight observation, Proposition 3.1, extracts ordinary formal codegrees from the Casimir spectrum even though the ring may be noncommutative. Known near-group, zero-spectrum and cyclotomic tests provide other general obstructions [24, 27, 20]. Standard quantum group, pointed, Tambara–Yamagami and representation categories supply the positive side [28, 40, 21, 41].

The accepted study of integral Grothendieck rings [18, 19] supplies a necessary divisibility test and explicit group-theoretical rank-eight models and exclusions. Being an integral or Drinfeld ring does not itself supply a fusion category, so only its explicit categorical constructions enter our positive list.

Theorem 1.1 (Certified rank-eight frontier)**.** *For the complete census through multiplicity two, the following trichotomy is exhaustive as an* analysis status*.*

Exact multiplicity	*Fusion rings*	*Excluded generally*	*Unitarily realized*	*Undecided generally*
1	96	45	48	3
2	877	801	24	52
Total	973	846	72	55

Every excluded ring is ruled out over $\mathbb{C}$ *with no pivotality assumption. Each positive model is matched to the entire tensor under a unit-preserving permutation, and is therefore a Grothendieck ring. For the* 55 *undecided rings, neither existence nor nonexistence is asserted. In particular, these numbers are* certified lower bounds *of* 48 *and* 24 *for the number of Grothendieck rings at the two multiplicities, rather than total counts of Grothendieck rings.*

The earliest undecided tensors are now `R8_1_9` and `R8_2_24`. The former first open ring `R8_1_3`, of global FP dimension 20, is realized below by a twisted group-theoretical category. Its dimension vector agrees with that of the commutative ring $\mathrm{Rep}(\mathrm{Dic}_5)$, although its multiplication is noncommutative. Thus full tensor comparison remains essential even after a suitable dimension profile is found.

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2. Trusted input and stable records

A rank-r fusion ring has a distinguished basis $B = (b_0 = 1, b_1, \dots, b_{r-1})$, structure constants $N_{ij}^k \in \mathbb{Z}_{\geq 0}$, and a duality involution $i \mapsto i^*$ satisfying

$$b_i b_j = \sum_k N_{ij}^k b_k, \qquad N_{ij}^0 = \delta_{j,i^*}, \qquad N_{ij}^k = N_{i^*k}^j = N_{kj^*}^i.$$

Its exact multiplicity is $\max_{i,j,k} N_{ij}^k$. A based isomorphism permutes the basis, fixes b_0, and preserves every structure constant; it automatically preserves duality. We do not identify tensors by character table, spectrum or FP dimension vector alone.

The source is the public *Fusion Ring Census* [11] at Git commit `b86eb6dc6eeeafe5fc5f6906d7b074e0322e9726`. It contains `results/rank8/tables.txt.gz` and `results/rank8/counts.csv`, with audit file `verification/rank8/audit.json`. The SHA-256 of the uncompressed tensor file is `97b86d2de8ca0352c5c6e14686cd1d43dbd52180dc53b2ff1ccf7a61788c9600`. The census verifies all duality types, associativity, nonnegative coefficients and uniqueness; its terminated-strata evidence justifies completeness through multiplicity two. We trust that source for counting and do not claim an independent exhaustive enumeration here.

Within each exact multiplicity m, sort full flattened $8 \times 8 \times 8$ tensors lexicographically and call the jth one $R8_m_j$. This is a deterministic local index, not an invariant of its original basis labelling. The files `data/decisions.json`, `data/realized_models.json` and `data/rank8_tables.txt.gz` give, respectively, one decision per input, the exact positive model tensors and the source multiplication tables. A SHA-256 of each JSON tensor prevents accidental source-record substitution. The comparison code minimizes the full tensor under all relabellings compatible with inexpensive intrinsic signatures; this is an isomorphism comparison, not a filter by numerical fingerprint.

3. Necessary criteria and their exact application

3.1. **Short tensor and subring tests.** If X is self-dual and $X^2 = 1 + kX$ with $k \geq 2$, its span with 1 is a fusion subring that cannot arise over $\mathbb{C}$ by [12, 22]. When exactly seven basis elements are invertible and one additional object X satisfies

$$gX = Xg = X, \qquad X^2 = \sum_{g \in G} g + \kappa X,$$

the rank-eight near-group case with $\kappa = 1$ or 2 is commutative and satisfies $0 < \kappa < |G| - 1$. Schopieray's general Proposition 3.2 [24] excludes it. The machine certificate checks the complete group and all displayed equations, not just the value of κ.

The zero-spectrum test is the finite nine-index condition of [27, Theorem 2.14] in the form reproduced in [16, Section 3]. It is a necessary condition for an associative tensor lift, with no pivotality assumption. Its C++ scanner searches the nine indices on the full tensor and the Python checker independently recomputes every coefficient, positivity condition and vanishing contraction. The native arrays are sized for rank eight, and the checker has no rank-six restriction.

Every duality-closed, multiplication-closed based subset of a category gives a fusion subcategory. We compare every such proper subring of ranks two through six against the *complete positive list* in the corresponding lower-rank theorem. The valid bounds are unrestricted for rank two and three, 128 for rank four, 20 for rank five, and 2 for rank six [12, 13, 14, 15, 16]; every rank-eight subring in this manuscript also has multiplicity at most two. A rank-seven subring is not excluded merely because it is absent from an incomplete rank-seven list.

The rank-seven analysis [17] provides four specific, fully checked negative multiplicity-two tensors that occur as rank-eight closed subrings. We transfer their exact signed-sphericalization witnesses by complete seven-dimensional based isomorphism and recheck all four proof trees. This separate

stage gives exclusions only for rings actually containing one of those four proved negative subrings; it makes no claim that the rank-seven positive catalogue is exhaustive.

The involution-module theorem of [16, Section 5] says that an invertible involution fixing basis elements forces a finite anti-invariant module representation. Where its stated hypothesis is met, the script searches *all* bounded integer matrices and verifies multiplication, transpose, parity and entry bounds. The FP dimension quotient of a hypothetical fusion subcategory must likewise be an algebraic integer [21, Proposition 8.15]; a rational coefficient in its exact minimal polynomial excludes categorification.

3.2. **A rank-eight noncommutative codegree lemma.** For any fusion ring, write

$$\Omega = \sum_i b_i b_{i^*}, \qquad C_R = \sum_i N_i N_i^{\mathsf{T}}.$$

For an irreducible representation E of $R \otimes \mathbb{C}$ of degree e_E, the central element Ω acts as the scalar $e_E f_E$, where f_E is its ordinary formal codegree. The regular representation contains E with multiplicity e_E, so that scalar has multiplicity e_E^2 in the characteristic polynomial of C_R [23, 16].

Proposition 3.1 (Noncommutative rank-eight codegrees)**.** *Let R be a noncommutative fusion ring of rank eight. Its semisimple complex algebra has Wedderburn decomposition $\mathbb{C}^4 \oplus M_2(\mathbb{C})$. The formal codegree of the unique matrix-block representation is a rational integer. Every irreducible factor of the characteristic polynomial of C_R of degree at least two consists entirely of formal codegrees of one-dimensional representations. Consequently, if any such factor fails Ostrik's d-number divisibility condition, R has no complex fusion categorification.*

Proof. The FP dimension character is one-dimensional, so at least one scalar summand exists. Noncommutativity requires a matrix summand of dimension at least two. Since the sum of the squares of the degrees of simple modules is eight, there can be only one matrix summand and it must have degree two; the four remaining dimensions are one. The rational based algebra is split by some number field. Galois automorphisms permute its irreducible representations preserving their degrees, so the unique two-dimensional representation is Galois-fixed. Its formal codegree is an algebraic integer by [23] and rational by Galois invariance, hence belongs to $\mathbb{Z}$. Its Casimir eigenvalue is therefore rational. Every nonlinear irreducible factor of the integral Casimir polynomial comes from the four one-dimensional representations; on each, the eigenvalue is exactly the formal codegree. Finally, formal codegrees of complex fusion categories are d-numbers [23]. For an irreducible polynomial $q(t) = t^d + a_1 t^{d-1} + \cdots + a_d$, this requires $|a_d|^j \mid a_j^d$ for every $j = 1, \ldots, d$. Failure of one exact integer divisibility gives the assertion. □

The commutative branch uses the ordinary d-number test on every Casimir factor. The script never treats raw Casimir eigenvalues as formal codegrees of noncommutative representations without Proposition 3.1. We also ran the rational-indivisibility screen of [25, Theorem 4.4], noncommuting field-automorphism witnesses from [16], and finite-field cyclotomic witnesses from [14]. These three screens add no exclusions in the present bound; their zero rows and exact code remain in the stage data. Schopieray's greedy formal-codegree obstruction $2, 3, 7$ [26] cannot occur before multiplicity five, since its forced rank-four subring has a coefficient five, so it is not an active step here.

3.3. **Integral divisibility and the complete signed quotient test.** The integral Drinfeld condition [18, Section 2.3] says that, if all simple FP dimensions of a categorifiable ring are integers, every ordinary formal codegree is a positive integer dividing its global FP dimension. For a commutative tensor, the verifier finds its unique positive FP character by an exact integral eigenvector of the strictly positive matrix $\sum_i N_i$. It then compares every exact linear factor of $\det(tI - C_R)$ with $D = \sum_i \mathrm{FPdim}(b_i)^2$. A nonintegral factor, or an integer codegree that does not divide D, is an unconditional complex no-go. This stage includes concrete multiplicity-two failures with $D = 156$ and 110, respectively, at $R8_2_288$ and $R8_2_392$.

The archived rank-eight appendix of [19] prints full tensors and explicit categorification decisions. Its items C.3.2(5) and C.3.2(7) are excluded by the paper's no-induction-matrix Lemmas 4.3 and 4.2. The appendix-to-census verifier checks all 8^3 entries under a unit-preserving permutation before transferring either published exclusion. We rely on those cited lemmas for the induction obstruction; their original Normaliz computation is not repeated here. The same transfer finds group-theoretical models in items C.3.2(3), C.3.2(6), and C.3.2(11). Item C.3.2(4) matches $R8_1_3$ but supplies no existence or exclusion claim; the explicit twisted group-theoretical construction below now realizes that ring.

Finally, the oriented signed-sphericalization theorem of [16, 17] applies even when the alleged category has no pivotal structure. The canonical sphericalization has two lifts of each simple; quotienting its Grothendieck ring by the negative central invertible yields an ordered signed multiplication tensor. We enumerate every cyclic and duality orbit of signed coefficients within the exact parity bounds, require all ordered associativity equations, then check every possible spherical character by the inherited norm, duality, and dimension-order criteria. A finite search that times out or has an unrefuted branch supplies no exclusion. The attached campaign records each attempted multiplicity-one tensor, incomplete result, and timeout separately; only certificates whose complete signed branches independently regenerate are admitted to the ledger.

3.4. **Grading obstructions with rank-one components.** The following short arguments give five further exclusions without any pivotality assumption. In a faithful grading $\mathcal{C} = \bigoplus_{g\in G} \mathcal{C}_g$, every component is an invertible bimodule category over $\mathcal{D} = \mathcal{C}_e$, and $g \mapsto [\mathcal{C}_g]$ is a homomorphism to $\mathrm{BrPic}(\mathcal{D})$ [7, Theorem 1.3 and Section 6].

Lemma 3.2. *If a pointed category* Vec_A^ω *has a module category with one simple object, then* $[\omega] = 0$.

Proof. Identify the module category with Vec. Each invertible object acts by the identity functor, up to isomorphism. The scalar module constraints form a nonzero 2-cochain whose coboundary is ω, by the module pentagon. Hence the associator class is trivial. □

Proposition 3.3. *The rings* $R8_1_26$*,* $R8_2_116$*, and* $R8_2_189$ *have no complex fusion categorification.*

Proof. In the original zero-based census labels, $R8_1_26$ has a faithful C_4 grading with components

$$\{0,3,4\}, \quad \{1\}, \quad \{5,6,7\}, \quad \{2\}.$$

Its neutral group is $A = C_3$. Lemma 3.2 identifies its neutral category with Vec_{C_3}. By [7, Corollary 1.2], its Brauer–Picard group is the orthogonal group of $A \oplus A^*$ with hyperbolic quadratic form. Over $\mathbb{F}_3$ this consists of $I, -I, J, -J$, where J exchanges the two coordinates. Every square is the identity, so $[\mathcal{C}_2] = [\mathcal{C}_1]^2 = 1$. On the other hand, the degree-two invertible object $s = 5$ conjugates $a = 3$ to $a^{-1} = 4$. Its component is the regular left module with right action twisted by inversion. It cannot be equivalent to the regular bimodule: such an equivalence would send s to some $b \in A$ and the relation $as = sa^{-1}$ would imply $ab = ba^{-1}$, which is false in C_3.

For $R8_2_116$ and $R8_2_189$, the universal grading has five components, of ranks $4, 1, 1, 1, 1$, so its group is C_5. Their neutral groups are respectively C_4 on $\{0, 1, 2, 7\}$ and C_2^2 on $\{0, 5, 6, 7\}$. Again their associator classes are trivial. In the first case the Brauer–Picard group embeds in $\mathrm{GL}_2(\mathbb{Z}/4)$, of order 96; in the second it is $O_4^+(\mathbb{F}_2)$, of order 72. Neither group has an element of order five. Hence the component homomorphism is trivial, forcing every component to be equivalent to the rank-four regular bimodule, contrary to the rank-one components. The verifier enumerates the small orthogonal groups and checks every grading support identity exactly. □

Proposition 3.4. *The rings* $R8_2_78$ *and* $R8_2_168$ *have no complex fusion categorification.*

Proof. Each has a faithful grading with a rank-one component and neutral fusion rules $\mathrm{Rep}(S_3)$: $g^2 = 1$, $gx = xg = x$, and $x^2 = 1 + g + x$. A rank-one module category gives a fiber functor

on the neutral category $\mathcal{D}$, so $\mathcal{D} \simeq \mathrm{Rep}(H)$ for a semisimple Hopf algebra of dimension six. The dimension-six classification [10] and the rank-three fusion rules force $H \simeq \mathbb{C}[S_3]$. The dual category of its fiber-functor module is $\mathrm{Rep}(H^*) \simeq \mathrm{Vec}_{S_3}$, of rank six [21, Example 7.12.26]. But invertibility of the homogeneous $\mathcal{D}$-bimodule would identify that dual with $\mathcal{D}$, of rank three, a contradiction. □

3.5. **Solvability, equivariantization and module ranks.** We give three further general exclusions using only small groups and exact fusion rules. By [3, Theorem 1.6], a fusion category of FP dimension $p^a q^b$ is solvable. The characterization in its Definition 1.2 and Proposition 4.4 expresses it as successive cyclic-prime extensions and equivariantizations. If its adjoint subcategory is the whole category, its last nontrivial step cannot be an extension. Thus it is an actual C_p-equivariantization, with a central $\mathrm{Rep}(C_p)$ whose underlying invertibles embed into the category. We use the orbit/stabilizer description of equivariant simples [6, Section 2]; cyclic stabilizers have only one-dimensional irreducible projective representations.

Proposition 3.5. *The ring R8_1_29 has no complex fusion categorification.*

Proof. Its invertibles form C_6, its other two simples X, Y are self-dual of dimension three, and its adjoint ring is the whole ring. Its dimension is 24, so a putative category $\mathcal{C}$ is $\mathcal{D}^{C_p}$ with $p = 2$ or 3.

For $p = 3$, the order-three character subgroup fixes X, Y and has two free orbits on the six invertibles. The forgetful images of X, Y split into three distinct invertibles each; those of the invertibles give two more. Thus $\mathcal{D} = \mathrm{Vec}_A^\omega$, $|A| = 8$, with two free C_3-orbits. The only groups of order eight admitting an order-three automorphism are C_2^3 and Q_8. On C_2^3 the action has a fixed line and an irreducible plane. The three nonzero elements in that plane form a free orbit whose product has coefficient two into the same free orbit. The output stabilizer is trivial, so associator and tensorator choices cannot change this coefficient. On Q_8, inversion exchanges the orbits $\{i, j, k\}$ and $\{-i, -j, -k\}$; their two induced dimension-three objects are distinct duals. These conclusions contradict multiplicity one and self-duality, respectively.

For $p = 2$, the character involution exchanges X, Y. De-equivariantization has the fusion rules

$$G(\mathcal{D}) = C_3 = \langle a \rangle, \qquad a\rho = \rho a = \rho, \qquad \rho^2 = 1 + a + a^2 + 2\rho.$$

It has dimension twelve and trivial universal grading. The same last-step argument gives $\mathcal{D} = (\mathrm{Vec}_V^\omega)^{C_3}$, where $V = C_2^2$ and C_3 permutes the three nonzero elements transitively. No assumption on ω is made. A monoidal automorphism of $\mathrm{Id}_\mathcal{D}$ has scalar one on ρ, because ρ occurs in ρ^2, and then on all invertibles. Consequently the map from central invertibles to underlying invertibles is injective, and the central $\mathrm{Rep}(C_3)$ is unique.

The C_2 action on $\mathcal{D}$ fixes every simple: exchanging a, a^2 would produce a dimension-two simple upstairs. Every such tensor autoequivalence preserves the unique central $\mathrm{Rep}(C_3)$. Its restriction is tensor-isomorphic to the identity, since $H^2(C_3, \mathbb{C}^\times) = 0$. Equivariantization/de-equivariantization [4, Theorem 4.18(i) and Proposition 4.19] therefore lifts it to a C_3-equivariant tensor autoequivalence of Vec_V^ω. Its object permutation centralizes C_3 in $\mathrm{Aut}(V) = S_3$, so lies in C_3. Composing with the inverse action functor removes that permutation without changing the descended equivalence class.

Identity-on-objects tensor autoequivalences of Vec_V^ω are classified by $H^2(V, \mathbb{C}^\times) = C_2$: the two associators cancel in the tensorator equation. Their equivariant structures have ambiguity $H^1(C_3, \widehat{V}) = 0$, since multiplication by three is invertible on $\widehat{V}$. Both possible classes have explicit representatives. Besides the identity, take

$$J(u, v) = (-1)^{u_1 v_1 + u_1 v_2 + u_2 v_2}, \qquad u, v \in \mathbb{F}_2^2.$$

This nontrivial normalized 2-cocycle is C_3-invariant and satisfies $J^2 = 1$. Thus it commutes even with nontrivial scalar tensorators of the existing C_3 action and descends with identity comparisons. There are no further C_2 coherence choices, because $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{D}) = 1$.

The identity action leaves the coefficient two in ρ^2. For the J action, all degree-zero channels have $J(u, u) = -1$ for $u \neq 0$; for each nonzero output degree, the two ordered channels have opposite

signs. Hence each lift $\rho_\pm$ has square containing the three negative-character invertibles and one copy of each $\rho_\pm$. The unit occurs in $\rho_+\rho_-$, not in either square. The two lifts are distinct duals, again contradicting the self-duality in R8_1_29. This exhausts both prime branches for arbitrary pointed associators. □

Lemma 3.6 (Module-rank divisibility)**.** *For a fusion category $\mathcal{B}$, let $g_\mathcal{B}$ be the greatest common divisor of the ranks of its indecomposable finite semisimple module categories. If $\mathcal{B} \subseteq \mathcal{D}$ is a fusion subcategory, then $g_\mathcal{B} \mid \mathrm{rank}(\mathcal{D})$.*

Proof. Decompose $\mathcal{D}$, with its left tensor action by $\mathcal{B}$, into indecomposable module categories. Their ranks add to $\mathrm{rank}(\mathcal{D})$. This statement uses ranks in the specified module presentation; it makes no assertion that module ranks are Morita invariant. □

Proposition 3.7. *The rings R8_1_45 and R8_1_47 have no complex fusion categorification.*

Proof. Each ring has simple dimensions $(1,1,2,3,3,4,4,4)$ and admits labels $1, h, E, P, Q, W_0, W_1, W_2$ with

$$h^2 = 1, \quad hE = E, \quad hP = Q, \quad hW_i = W_i, \quad P^* = Q,$$
$$E^2 = 1 + h + E, \qquad EP = EQ = P + Q, \qquad PQ = 1 + E + P + Q.$$

The adjoint ring is the whole ring. Dimension 72 and the invertible group C_2 therefore force a putative categorification to be $\mathcal{C} = \mathcal{D}^{C_2}$ with $\mathrm{FPdim}(\mathcal{D}) = 36$. The forgetful images are

$$F(1) = F(h) = 1, \quad F(E) = a + a^2, \quad F(P) = F(Q) = Y, \quad F(W_i) = U_i + V_i.$$

Here a has order three, Y is simple of dimension three, and the six U_i, V_i are distinct simples of dimension two. Thus $\mathrm{rank}(\mathcal{D}) = 10$. The full fusion subcategory $\mathcal{B}$ on $1, a, a^2, Y$ satisfies

$$aY = Ya = Y, \qquad Y^* = Y, \qquad Y^2 = 1 + a + a^2 + 2Y.$$

The C_2 action exchanges a, a^2 and fixes Y; its equivariantization inside $\mathcal{C}$ has the displayed nonselfdual P, Q.

As in the preceding proof, solvability and trivial universal grading imply $\mathcal{B} = (\mathrm{Vec}_V^\omega)^{C_3}$, $V = C_2^2$, with the transitive action. The restriction map

$$H^3(V, \mathbb{C}^\times) \longrightarrow \prod_{\substack{L \le V \\ |L|=2}} H^3(L, \mathbb{C}^\times)$$

is an isomorphism $C_2^3 \to C_2^3$. In integral-cohomology generators u^2, v^2, uv, the three restrictions of (r,s,t) are $(r, s, r+s+t)$. Since C_3 permutes the factors, there are exactly two invariant classes: zero and the class nontrivial on every subgroup of order two.

Suppose first $[\omega] = 0$. The lift of the object C_3 action is unique in $\mathrm{Aut}_\otimes(\mathrm{Vec}_V) = C_2 \times S_3$, and its coherence is unique because $H^2(C_3, \widehat{V}) = 0$. Therefore $\mathcal{B} \simeq \mathrm{Rep}(A_4)$. Its central $\mathrm{Rep}(C_3)$ is unique, since $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{B}) = 1$, and is preserved by the C_2 action. Sequential equivariantization and de-equivariantization [3, Section 2.6 and Proposition 2.10][4, Proposition 4.19] identify $\mathcal{B}^{C_2}$ with $(\mathrm{Vec}_V)^{S_3}$. The action on nonzero elements is the usual faithful S_3 permutation action; inversion on the character C_3 forces the group $C_3 \rtimes C_2 = S_3$.

There are exactly two lifts of this object action to $C_2 \times S_3$, given by the trivial or parity homomorphism into $H^2(V, \mathbb{C}^\times) = C_2$. Their coherence ambiguity vanishes: $H^j(S_3, \widehat{V}) = 0$ for $j > 0$, by the normal C_3 subgroup, whose positive cohomology and fixed space on $\widehat{V}$ both vanish. Both lifts yield self-dual dimension-three orbit objects. For completeness choose a stabilizing transposition τ swapping coordinates and its fixed nonzero element $v = (1,1)$. For the trivial lift, the equivariance scalars $u = \pm 1$ act trivially on the unit channel of $v \otimes v$. For the other lift choose tensorator $\mu(a,b) = (-1)^{a_1 b_2}$. Its square is trivialized by $\eta_a = (-1)^{a_1 a_2}$, so $\eta_v = \mu(v,v) = -1$. The equivariance scalar satisfies $u^2 = \eta_v$ and acts on the unit channel by $u^2/\mu(v,v) = 1$. A τ-invariant character

change of η is one on v, so cannot change the conclusion. Frobenius reciprocity for the orbit induction gives a tensor unit in each of the two squares. This contradicts $P^* = Q$.

It remains that $[\omega]$ is nontrivial on every order-two subgroup. The crossed product of Vec_V^ω by C_3 is pointed, and realizes $\mathcal{B}$ as the group-theoretical category

$$\mathcal{B} \simeq \mathcal{C}(A_4, \Omega, K, \psi), \qquad K = C_3, \qquad [\Omega]|_V = [\omega], \quad [\Omega]|_K = 0.$$

By the classification of pointed module categories and its double-coset rank formula [5, Example 2.1, Theorem 3.1 and Proposition 3.1], indecomposable $\mathcal{B}$-module categories correspond to subgroups $H \leq A_4$ with $[\Omega]|_H = 0$ and permissible cochains. If $|H|$ were even, Cauchy's theorem would give a subgroup of order two, contradicting the restriction of Ω. Thus H is trivial or has order three. For $H = 1$, the four double cosets in $K\backslash A_4/H$ have trivial intersections, giving module rank four. For $H = C_3$, conjugate it to K. There are two double cosets, with intersections C_3 and 1; projective cocycles on C_3 are coboundaries, so their contributions are three and one. Every indecomposable $\mathcal{B}$-module category therefore has rank four, regardless of the permissible cochains. Lemma 3.6 would force $4 \mid \mathrm{rank}(\mathcal{D}) = 10$, a contradiction. □

The portable verifiers check the source-bound tensors, the displayed fusion identities, the complete small cocycle identities, and all relevant A_4 subgroups and double cosets. These finite checks certify the hypotheses; the categorical exhaustion is supplied by the proofs above. Neither argument assumes a braiding or a pivotal structure.

3.6. **Rank-one modules and semisimple Hopf algebras.**

Proposition 3.8. *The rings* R8_2_41*,* R8_2_86*,* R8_2_162*,* R8_2_140 *and* R8_2_784 *have no complex fusion categorification.*

Proof. For the first three rings, a faithful C_2 grading has a rank-one nonneutral component and a dimension-sixteen neutral category $\mathcal{D}$ of type $(1^4, 2^3)$. A rank-one module supplies a fiber functor, so $\mathcal{D} = \mathrm{Rep}(H)$ for a semisimple Hopf algebra of dimension sixteen. Invertibility of the component further requires $\mathrm{Rep}(H^*) \simeq \mathcal{D}$; in particular both groups of grouplikes have order four. Kashina's classification [43, Theorem 1.2 and Table 1] leaves only $H_{C:\sigma_1}$, whose representation ring is $\mathrm{Rep}(D_{16})$: its invertibles form C_2^2, and all three dimension-two simples are self-dual. The rings R8_2_41, R8_2_86 instead have cyclic invertibles, while R8_2_162 has a nonselfdual dimension-two pair.

For R8_2_140, census indices $\{0, 1, 2, 3, 7\}$ give the neutral Tambara–Yamagami category on C_4, of dimension eight. Each other simple separately gives a rank-one left module: an invertible acts as one and the dimension-two object acts as two. Thus the neutral category has a fiber functor and is $\mathrm{Rep}(H)$ with $G(H^*) = C_4$. Kashina's prime-power theorem [43, Theorem 1.1] says that a semisimple Hopf algebra of prime-power dimension with cyclic grouplikes is a group algebra. It would give $H^* \simeq \mathbb{C}[C_8]$, which has eight grouplikes, a contradiction. This argument does not require the rank-one modules to be grading components.

For R8_2_784, the neutral component has dimension twenty-two, rank seven and type $(1^2, 2^5)$; the other component has a unique simple of dimension $\sqrt{22}$. Its fiber functor reconstructs a Hopf algebra of dimension $2 \cdot 11$. The Hopf pq classification [10] and the nonpointed neutral rules force $\mathcal{D} = \mathrm{Rep}(D_{22})$. Its fiber-functor dual is $\mathrm{Vec}_{D_{22}}$, of rank twenty-two. The invertible grading component would identify this dual with $\mathcal{D}$, of rank seven. This is impossible. □

3.7. **Preserved Tannakian subcategories.**

Proposition 3.9. *The rings* R8_2_697*,* R8_2_808 *and* R8_2_181 *have no complex fusion categorification.*

Proof. For R8_2_697, dimension sixty-eight, type $(1^4, 4^4)$ and trivial universal grading, with all four invertibles fixing each noninvertible, force a C_2 equivariantization with descent $\mathcal{D}$ of dimension thirty-four and type $(1^2, 2^8)$. Its adjoint is the whole category: a proper adjoint has dimension $1, 2$ or 17; a

prime-dimensional adjoint is pointed, but these possibilities cannot contain the required dimension-four squares with only two invertibles. Solvability and prime descent now give $\mathcal{D} = (\mathrm{Vec}^{\omega}_{C_{17}})^{C_2}$, with inversion. Since $U(\mathcal{D}) = 1$, central lifts of its invertibles are unique. The outer involution therefore preserves its central $\mathrm{Rep}(C_2)$, and the combined equivariantization group is C_2^2, as its four linear characters are the census invertibles. Every $C_2^2 \to \mathrm{Aut}(C_{17})$ has image of order at most two. A nonzero orbit has size two and cyclic stabilizer; it therefore supplies dimension-two equivariant simples. The target has only dimension-four noninvertibles.

For $R8_2_808$, dimension $186 = 2 \cdot 3 \cdot 31$ and the integral *pqr* theorem make any categorification group-theoretical [3, Theorem 9.2]. Groups of order 186 are solvable, so the category is solvable. The adjoint, type $(1^2, 2, 6^5)$, and the character involution fixing all noninvertibles force C_2 descent of type $(1^3, 3^{10})$ and dimension ninety-three. The same prime-dimensional-adjoint argument shows that its universal grading is trivial. Thus it is $(\mathrm{Vec}^{\omega}_{C_{31}})^{C_3}$ with nontrivial order-three action. Uniqueness of the central C_3 preserves it under the outer involution. The combined group is S_3, because its embedded representation subcategory has type $(1^2, 2)$. Every homomorphism $S_3 \to \mathrm{Aut}(C_{31})$ kills its normal C_3, contradicting that action.

For the commutative ring $R8_2_181$, the neutral category has dimension twenty-one, type $(1^3, 3^2)$ and trivial universal grading. Solvability makes it $(\mathrm{Vec}^{\omega}_{C_7})^{C_3}$, with multiplication by two or four on C_7. The action on $H^3(C_7, \mathbb{C}^\times) = C_7$ is multiplication by four or two, so its invariant associator class is zero. Coprime action cohomology then identifies the neutral category with $\mathrm{Rep}(C_7 \rtimes C_3)$. Its central C_3 has unique lifts. The three odd simples have free, identical left and right C_3 actions, so the odd bimodule fixes these central lifts pointwise. Relative-center equivariantization [46] and $H^2(C_2, \mathbb{C}^\times) = 0$ give six central invertible lifts upstairs. The unique order-three subgroup has trivial self-braiding, inherited from the neutral $\mathrm{Rep}(C_3)$, and is an embedded Tannakian subcategory. Its de-equivariantization has the Tambara–Yamagami rules on C_7. The induced C_3 symmetry must preserve its nondegenerate symmetric bicharacter $\chi(a, b) = \exp(2\pi i k a b/7)$, $k \neq 0$. Multiplication by two or four cannot do so: neither square is one modulo seven. This proves the contradiction without a pivotal or a fermionic-transparency assumption. □

3.8. **A dimension-twenty-eight adjoint criterion.**

Lemma 3.10. *A nonpointed integral fusion category of dimension twenty-eight has universal grading C_2 and adjoint fusion ring* $\mathrm{Rep}(D_{14})$.

Proof. Jordan–Larson's theorem [44, Theorem 1.1] applies with $p = 7, q = 2$ and $7 \nmid q + 1$. Write the category as $\mathcal{C}(G, \omega, H, \psi)$, where $|G| = 28$. Sylow theory gives $G = C_7 \rtimes P$, with $P = C_4$ or C_2^2; the action is trivial or inversion. The possible subgroup orders are $1, 2, 4, 7, 14, 28$. For orders one and seven, or a central subgroup of order two, the cyclic stabilizers give a pointed category. For a noncentral subgroup H of order two, its normal closure is $C_7 \rtimes H = D_{14}$, of index two; support gives a faithful C_2 grading. For $|H| = 14$, H is normal and support again gives such a grading. An abelian $H = C_{14}$ would give a pointed category; otherwise $H = D_{14}$, whose Schur multiplier is trivial. For $H = G$ one obtains $\mathrm{Rep}(G)$; a nonabelian group of order twenty-eight has center C_2 and quotient D_{14}. In each of these nonpointed cases the neutral fusion ring is $\mathrm{Rep}(D_{14})$.

It remains to justify this ring identification for $|H| = 4$, and also the cocycle reduction used for a reflection complement above. Conjugate H to P. The positive-degree cohomology of C_7 is seven-primary, so the coprime Lyndon–Hochschild–Serre sequence has no mixed positive-degree terms. Since $\omega|_P = d\psi$, the associator class is represented by a P-invariant class on C_7. Choose a normalized μ_7 representative, average over P, and raise to the inverse of $|P|$ modulo seven. This gives a strictly invariant representative α. On $C_7 \rtimes P$ use

$$\omega_\alpha((n_1, p_1), (n_2, p_2), (n_3, p_3)) = \alpha(n_1, p_1 n_2, p_1 p_2 n_3).$$

It is identically one on P. The remaining 2-cocycle ψ is removed by its inflation along $G \to P$. Thus the category is $(\mathrm{Vec}^{\alpha}_{C_7})^P$ with strict permutation action. A trivial action is pointed. Otherwise its

kernel C_2 acts strictly trivially and its characters grade the category, with neutral $(\mathrm{Vec}_{C_7}^{\alpha})^{C_2}$. This has the $\mathrm{Rep}(D_{14})$ fusion ring; the same argument with a reflection complement supplies the previous order-two case.

Finally this neutral ring is adjoint to itself. For its three dimension-two objects, indexed by nonzero elements modulo inversion, $X_i^2 = 1 + \varepsilon + X_{2i}$, and doubling cycles the three indices. The adjoint of the full category is contained in the displayed neutral component and contains its adjoint, so it equals that component. Consequently the universal grading is exactly C_2. □

Proposition 3.11. *There is no fusion category with dimension fifty-six, type* $(1^4, 2, 4^3)$ *and trivial universal grading. In particular* R8_2_694 *is generally excluded.*

Proof. Solvability and trivial universal grading force a last equivariantization. A C_7 equivariantization would supply seven invertibles, so the last group is C_2. Its descent $\mathcal{D}$ has dimension twenty-eight and is integral by the orbit/stabilizer dimension formula. It cannot be pointed, since a C_2 equivariantization of a pointed category has all dimensions at most two. Lemma 3.10 therefore applies. The outer involution preserves $\mathcal{D}_{\mathrm{ad}}$ and permutes its three dimension-two simples, fixing at least one. Its cyclic stabilizer has trivial projective multiplier, so its two characters produce two distinct dimension-two equivariant simples. The proposed category has only one. This is a contradiction. □

3.9. **A two-object orbit obstruction in dimension ninety-six.**

Proposition 3.12. *A fusion category of type* $(1^4, 2, 4, 6^2)$ *has a nontrivial universal grading. In particular* R8_2_404*,* R8_2_418*,* R8_2_604*,* R8_2_692 *have no complex fusion categorification.*

Proof. Suppose such a category has trivial universal grading. Its dimension is ninety-six, and its invertible group has order four. Solvability forces a last C_2 equivariantization $\mathcal{C} = \mathcal{D}^{C_2}$. Whichever nontrivial character occurs, it fixes the unique dimension-two and dimension-four simples. Thus the outer involution exchanges the two dimension-two simples of $\mathcal{D}$. Depending on whether it fixes or exchanges the dimension-six pair upstairs, $\mathcal{D}$ has type $(1^4, 2^2, 3^4)$ or $(1^4, 2^2, 6)$ and dimension forty-eight.

Its canonical adjoint is preserved by the outer involution. A proper adjoint's dimension therefore has the form $a + 4b + 9c + 36e$, where $a \in \{1, 2, 4\}$, $b \in \{0, 2\}$, c is even, and e is zero or one according to the displayed type. The proper divisors of forty-eight among these possibilities are only $1, 2, 4, 12$. A pointed adjoint of dimension at most four cannot contain the square of a dimension-three or dimension-six simple, since each invertible occurs there with coefficient at most one. An adjoint of dimension twelve consists of all four invertibles and both dimension-two simples. Every other grading component would then have dimension twelve using only dimension-three or dimension-six simples; a sum of their squared dimensions is divisible by nine, so this is impossible. Hence $\mathcal{D}_{\mathrm{ad}} = \mathcal{D}$.

A second prime descent gives $\mathcal{D} = \mathcal{E}^{C_2}$, $\mathrm{FPdim}(\mathcal{E}) = 24$. The complete cyclic-orbit dimension partitions are $(1^6, 3^2)$ and $(1^2, 2, 3^2)$. In the first case the dimension-two pair of $\mathcal{D}$ comes from two free orbits in the invertible group A of order six. The inner involution fixes exactly two elements of A. For $A = C_6$ it is inversion, and for $A = S_3$ it is conjugation by a transposition. In either group the coefficients of the two induced objects in their own squares are respectively zero and one, by ordinary orbit addition. The output stabilizers are trivial, so associator or tensorator phases cannot change these coefficients. The outer involution cannot exchange this pair as required.

In the second case the unique dimension-two simple x of $\mathcal{E}$ is fixed by the inner involution, with $x^2 = 1 + g + x$. Its two equivariant lifts x_+, x_- are exchanged by the central character ε. Therefore their squares are equal, because $\varepsilon^2 = 1$ and ε is central. Each square has precisely one constituent from the dimension-two pair, by forgetting to the single x channel. The outer involution exchanges x_+, x_- but preserves their common square. Its unique dimension-two constituent would have to be fixed, which is impossible. Both dimension partitions and every initial character choice have therefore been excluded. The supplementary check binds the full four source tensors and exhausts these small dimension and orbit cases. □

3.10. **An order-four action on rank-one components.**

Proposition 3.13. *The ring R8_2_68 has no complex fusion categorification.*

Proof. Its exact type is $(1^4, 4^2, 6^2)$, its invertibles form C_4, its adjoint is the whole ring, and its order-two invertible fixes all four noninvertibles. Solvability in dimension 108 therefore forces $\mathcal{C} = \mathcal{D}^{C_2}$ with $\mathcal{D}$ integral of dimension 54 and type $(1^2, 2^4, 3^4)$. The outer involution exchanges both noninvertible dimension classes in pairs and fixes the invertibles. The possible proper dimensions of its invariant adjoint are $1, 2, 9, 18, 27$. Pointed dimensions one and two cannot contain the square of a dimension-three object; dimensions nine and twenty-seven would contain dimension-two objects, contradicting strong Frobenius divisibility [3]. Thus either $\mathcal{D}_{\mathrm{ad}} = \mathcal{D}$ or its adjoint $\mathcal{E}$ has type $(1^2, 2^4)$ and dimension eighteen.

In the first case prime descent gives $\mathcal{D} = \mathcal{F}^{C_2}$, with $\mathcal{F}$ of type $(1^9, 3^2)$. In the second case $U(\mathcal{E}) = 1$: a grading of order two would have a dimension-nine neutral containing a dimension-two simple, again contradicting strong Frobenius; order-three components of dimension six would each require both invertibles; higher grading orders cannot accommodate a dimension-two simple. Hence $\mathcal{E} = (\mathrm{Vec}_A^{\omega})^{C_2}$, $|A| = 9$, with inversion on A. Its unique nontrivial central invertible g is the canonical Tannakian lift. The relative-center C_3 action of the grading of $\mathcal{D}$ fixes this lift. Since $H^2(C_3, \mathbb{C}^{\times}) = 0$, its central equivariant lifts, together with the unit lifts, form an abelian group of order six. The unique order-two lift restricts to g and inherits its self-braiding $+1$; it therefore gives a central $\mathrm{Rep}(C_2)$ in $\mathcal{D}$. This explicitly verifies bosonicity, rather than inferring it from centrality alone. De-equivariantization again gives $\mathcal{F}$ of type $(1^9, 3^2)$: a dimension-three object fixed by g would descend to dimension $3/2$, which is not an algebraic integer.

In both cases the order-two central lift in $\mathcal{D}$ is unique: its kernel is trivial when $U(\mathcal{D}) = 1$, and has order three when $U(\mathcal{D}) = C_3$. The outer involution preserves this Tannakian subcategory. Combining the two equivariantizations gives $\mathcal{C} = \mathcal{F}^{C_4}$, because the intermediate symmetric subcategory has dimension four and its four characters form the census C_4. Let β generate this coherent action. Its square acts by inversion on $A = \mathrm{Inv}(\mathcal{F})$ and fixes the two dimension-three objects.

The adjoint of $\mathcal{F}$ is Vec_A^{ω}, with a C_3 grading of ranks $9, 1, 1$. Rank-one components force $[\omega] = 0$ by their module pentagon. The action of β exchanges these two components; otherwise the original $\mathcal{C}$ would inherit a nontrivial C_3 grading. For $A = C_9$ no automorphism squares to inversion. For $A = C_3^2$, its matrix B satisfies $B^2 = -I$, so its minimal polynomial is the irreducible $t^2 + 1$ over $\mathbb{F}_3$ and $\det(B) = 1$. Its tensorator class $\eta \in H^2(A, \mathbb{C}^{\times}) = C_3$ is fixed by B; order-four coherence gives $4\eta = 0$, hence $\eta = 0$. Such a tensor autoequivalence fixes every rank-one left-module class, since those classes are $H^2(A, \mathbb{C}^{\times})$. But the two nonneutral components have different left-module classes: an equivalence, tensored with the inverse of one invertible bimodule, would identify the rank-nine regular module with a rank-one component. Thus β cannot exchange them, a contradiction. The supplement checks every source coefficient, adjoint dimension partition and matrix calculation; the central lift and action tower use [46, 4, 3]. □

3.11. **Ordered exclusion ledger and the first open cases.** An excluded tensor is removed after the first verified certificate in the following order. For each row, "left" still counts rings subsequently recognized as positive, so this is an exclusion table rather than a table of outstanding questions.

Criterion	New $m=1$	New $m=2$	Left $m=1$	Left $m=2$
Near-group stabilizer	1	1	95	876
Rank-two closed subring	0	10	95	866
Zero-spectrum tensor condition	7	80	88	786
Proved rank 2–6 subring	3	121	85	665
Four proved rank-seven subrings	0	12	85	653
Commutative codegree d-number	23	462	62	191
Rank-eight noncommutative codegree	1	17	61	174
Integral formal-codegree divisibility	1	6	60	168
Published no-induction-matrix lemma	2	0	58	168
Involution-module obstruction	0	2	58	166
Field-automorphism obstruction	0	0	58	166
Fusion-subcategory dimension quotient	0	1	58	165
Primitive formal-codegree test	0	0	58	165
Cyclotomic eigenvalue test	0	0	58	165
Complete oriented signed quotient	3	71	55	94
Graded Brauer–Picard obstruction	1	2	54	92
Rank-one fiber-functor obstruction	0	2	54	90
Solvability and equivariantization	3	0	51	90
Prime descent and Hopf obstructions	0	9	51	81
Dimension-96 two-object orbit	0	4	51	77
Dimension-108 order-four action	0	1	51	76

The first unresolved IDs are $R8_1_9$ and $R8_2_24$. Every negative stage is a theorem applied to tensor data or an exact based-tensor transfer of a published and proved no-go. Proposition 3.1 and the integral-divisibility stage both have certified first applications already at multiplicity one; the latter also resolves two multiplicity-two cases remaining open after the prior stages. These *first appearances* refer only to the two multiplicities of the complete rank-eight census. Nothing in this calculation establishes a multiplicity from which no new criterion is needed.

4. Exact positive models

Each standard family in this section admits a unitary complex realization. A pointed group ring of order eight is $K_0(\mathrm{Vec}_G)$ for the group reconstructed from its tensor. The five groups are C_8, $C_4 \times C_2$, C_2^3, D_8 and Q_8. The near-group ring $C_7 + 0$ is realized by a Tambara–Yamagami category [40]. If A is a rank-four unitary model from [14, Table 1] and B is either Vec_{C_2} or Fibonacci, the Deligne product $A \boxtimes B$ is a unitary rank-eight model with multiplication tensor $N_A \otimes N_B$. Its multiplicity is at most two precisely for the displayed matches.

The three archived models $\mathcal{C}(F_5, 1, D_5, 1)$, $\mathcal{C}(S_4, 1, A_4, 1)$, and $\mathcal{C}(\mathrm{PSL}(2,7), 1, C_7 \rtimes C_3, 1)$ are untwisted group-theoretical fusion categories, hence admit unitary realizations. Their exact appendix tensors are matched against the census in all 512 positions, with a stored basis permutation for every match [19, Appendix C.3.2(3), (6), (11)]. Some admit a second subgroup construction, also recorded in the model catalogue.

Proposition 4.1 (Representation and quantum-group templates)**.** *All three character-ring templates* $\mathrm{Rep}(D_{20})$*,* $\mathrm{Rep}(D_{26})$*, and* $\mathrm{Rep}(\mathrm{Dic}_5)$ *have rank eight and multiplicity one. The standard unitary* $SU(2)_7$ *category, and the even subcategories of* $SU(2)_{14}$ *and* $SU(2)_{15}$*, also have rank eight and multiplicity one. Their generated multiplication tensors occur in the supplied census and are matched by full based-ring isomorphisms.*

Proof. For $SU(2)_k$, the simples are the labels $0, \dots, k$ and

$$N_{ab}^{c} = 1 \quad \Longleftrightarrow \quad |a-b| \le c \le \min(a+b, 2k-a-b), \qquad a+b-c \equiv 0 \pmod 2;$$

all other coefficients are zero. The even labels close under multiplication and yield eight objects for $k = 14, 15$, while $k = 7$ already has eight labels [28]. The explicit tensor generator implements this formula directly.

For the dihedral group of order $2n$, take $n = 10$ or 13. Its two-dimensional characters ρ_j obey $\rho_i\rho_j = \rho_{i+j} + \rho_{|i-j|}$, with $\rho_0 = 1 + \beta$ and $\rho_{n-j} = \rho_j$. For $n = 10$, $\rho_5 = \alpha + \alpha\beta$, the four linear characters form C_2^2, $\beta\rho_j = \rho_j$ and $\alpha\rho_j = \rho_{5-j}$. For $n = 13$ there are two linear characters, with $\beta\rho_j = \rho_j$. Thus their ranks are respectively $4 + 4$ and $2 + 6$. For Dic_5 of order twenty the four linear characters χ_i form C_4, with

$$\rho_0 = \chi_0 + \chi_2, \quad \rho_5 = \chi_1 + \chi_3, \quad \rho_{10-j} = \rho_j, \quad \chi_i\rho_j = \begin{cases} \rho_j & i \text{ even}, \\ \rho_{5-j} & i \text{ odd}; \end{cases}$$

the same product $\rho_i\rho_j = \rho_{i+j} + \rho_{|i-j|}$ completes the ring. These are ordinary finite-group character identities [41]. The supplied generator expands each product and compares all 8^3 ordered coefficients after unit-fixing relabelling. A match is a ring isomorphism and the corresponding representation category supplies the construction. □

Multiplicity	Ring ID	One exact unitary model
1	`R8_1_1`	`Rep(Dic5)`
1	`R8_1_3`	$\mathcal{C}(C_5 \rtimes C_4, \omega_2, C_2, 1)$
1	`R8_1_4`	`Vec(G), |G|=8`
1	`R8_1_5`	`Vec(C4) x Fib`
1	`R8_1_6`	Bosonic quotient of $PSU(2)_6 \boxtimes \mathcal{C}(C_4, q)$
1	`R8_1_7`	Central square-root extension of pseudo-$PSU(2)_6$
1	`R8_1_10`	`Vec(G), |G|=8`
1	`R8_1_11`	Central-twisted flip extension of pseudo-$PSU(2)_6$
1	`R8_1_12`	Isotropic orbifold of $NG(C_{16}, 16)$
1	`R8_1_13`	`Vec(G), |G|=8`
1	`R8_1_14`	Central-twisted flip extension of $PSU(2)_6$
1	`R8_1_15`	$\mathrm{HI}(C_4)$
1	`R8_1_16`	`Vec(G), |G|=8`
1	`R8_1_19`	$(\mathrm{Fib} \boxtimes \mathrm{Fib}) \rtimes C_2$
1	`R8_1_20`	`TY(C7)`
1	`R8_1_22`	Diagonal parity in $\mathrm{TY}(C_3) \boxtimes \mathrm{Vec}_{C_4}$
1	`R8_1_23`	$\mathrm{Vec}_{Q_8}^{C_3}$
1	`R8_1_24`	`C(F5,1,D5,1)`
1	`R8_1_25`	$PSU(2)_6 \rtimes C_2$ with the flip action
1	`R8_1_27`	`TY(C3) x Vec(C2)`
1	`R8_1_30`	$\mathrm{TY}(C_6)^{C_2}$ with odd coherence -1
1	`R8_1_32`	`pseudo-PSU(2)_6 x Vec(C2)`
1	`R8_1_34`	$\mathrm{TY}(C_9)^{C_2}$ with odd coherence -1
1	`R8_1_35`	$\mathrm{TY}(C_3^2)^{C_2}$ with odd coherence -1
1	`R8_1_36`	`TY(C3) crossed C2`
1	`R8_1_39`	`TY(C3) x Fib`
1	`R8_1_42`	Pseudo-$PSU(2)_6 \rtimes C_2$ with the flip action
1	`R8_1_43`	`pseudo-PSU(2)_6 x Fib`
1	`R8_1_53`	Isotropic orbifold of $NG(C_4^2, 16)$
1	`R8_1_59`	`Rep(D10) x Vec(C2)`
1	`R8_1_61`	`One-half SO(12)_2`
1	`R8_1_65`	`Vec(G), |G|=8`
1	`R8_1_66`	`Vec(C2xC2) x Fib`
1	`R8_1_67`	`PSU(2)_6 x Vec(C2)`
1	`R8_1_69`	$\mathrm{HI}(C_2^2)$

Multiplicity	Ring ID	One exact unitary model
1	R8_1_70	Fib x Vec(C2) x Fib
1	R8_1_71	PSU(2)_6 x Fib
1	R8_1_72	PSU(2)_7 x Vec(C2)
1	R8_1_73	PSU(2)_14
1	R8_1_78	SO(9)_2
1	R8_1_79	Z(Vec(S3))
1	R8_1_83	Integral part of the $m = 0$ triality gauging
1	R8_1_85	Neutral part of S3-gauged SO(8)_1
1	R8_1_91	Rep(D_{26})
1	R8_1_92	PSU(2)_15
1	R8_1_93	Rep(D10) x Fib
1	R8_1_94	PSU(2)_7 x Fib
1	R8_1_95	Fib x Fib x Fib
2	R8_2_39	Asaeda–Haagerup AH_4
2	R8_2_143	$\mathrm{Vec}_{C_{17}}^{C_4}$
2	R8_2_154	$\mathrm{TY}(C_2^2)^{S_3}$ with parity coherence
2	R8_2_174	Rep(A4) x Vec(C2)
2	R8_2_179	$\mathrm{Vec}_{C_2^4}^{C_5}$
2	R8_2_202	C(S4,1,A4,1)
2	R8_2_212	Rep(A4) x Fib
2	R8_2_237	$\mathrm{Rep}(A\Gamma L(1,8))$
2	R8_2_239	C_3-algebra Morita dual of an S_3 extension of Vec_{C_7}
2	R8_2_241	$\mathcal{C}(\mathrm{PSL}(2,7), 1, C_7 \rtimes C_3, 1)$
2	R8_2_244	$\mathrm{Vec}_{C_2^4}^{C_3}$
2	R8_2_289	$\mathrm{Vec}_{C_2^4}^{C_3}$
2	R8_2_290	Charge-conjugation defects of $PSU(3)_4$
2	R8_2_313	Diagonal bosonic quotient of the two pseudo-$PSU(2)_6$ categories
2	R8_2_477	even 2D2 x Vec(C2)
2	R8_2_484	Asaeda–Haagerup AH_5
2	R8_2_491	even 2D2 x Fib
2	R8_2_493	Haagerup H1 x Vec(C2)
2	R8_2_557	Diagonal bosonic quotient of $PSU(2)_6 \boxtimes PSU(2)_6$
2	R8_2_575	Davydov–Runkel extension of $\mathrm{Rep}(H_{C:\sigma_1})$
2	R8_2_611	$\mathrm{TY}(C_2^2)^{S_3}$ with strict action
2	R8_2_839	Haagerup H1 x Fib
2	R8_2_855	Even D_{14} category
2	R8_2_868	Dual Haagerup–Izumi C_7 category

Five further multiplicity-one models are imported from exact AnyonWiki tensor matches: $R8_1_36$ is the crossed product of $\mathrm{TY}(C_3)$ by the order-two inversion action; $R8_1_61$ is the rank-eight fusion subcategory called one-half $SO(12)_2$; $R8_1_78$ is $SO(9)_2$; $R8_1_79$ is $Z(\mathrm{Vec}_{S_3}) \simeq \mathrm{Rep}(D(S_3))$; and $R8_1_85$ is the neutral component of the universal $\mathbb{Z}_2$ grading after gauging the full S_3 symmetry of $SO(8)_1$ [1, 2]. For the crossed product, the standard bicharacter $\chi(a,b) = \exp(2\pi iab/3)$ and positive $\tau = 1/\sqrt{3}$ are invariant under simultaneous inversion. The induced tensor action squares to the identity, and the crossed product is unitary. Its independently constructed tensor is matched in all 512 positions. The other four are standard unitary categories or fusion subcategories thereof; their complete sourced tensors and unit-preserving permutations are included in the positive catalogue. Existence of these models proves general complex categorifiability. No negative conclusion is imported merely from an AnyonWiki flag: the previously reported pentagon no-go for $R8_1_26$ is now replaced by the independent theoretical proof in Proposition 3.3.

4.1. **Twenty-two further constructions from extensions and subfactors.** The following constructions are verified by generating their multiplication tensors directly and comparing all 512 coefficients. No associator search is needed.

For $G = C_4$ and $G = C_2^2$, write $X_g = g\rho$ and use

$$gh = g + h, \qquad gX_h = X_{g+h}, \qquad X_g h = X_{g-h}, \qquad X_g X_h = (g - h) + m \sum_{a \in G} X_a.$$

At $m = 1$, Izumi's unitary generalized Haagerup constructions [8, Section 9] give $R8_1_15$ and $R8_1_69$, respectively. At $m = 2$, these are the Asaeda–Haagerup rings AH_4 and AH_5, giving $R8_2_39$ and $R8_2_484$. The C_4 construction is [9, Theorem 4.9]; existence of the C_2^2 category follows from its Lemma 3.15, Theorem 3.17 and Section 4. These are unitary subfactor categories.

The ring $R8_1_19$ is $(\mathrm{Fib} \boxtimes \mathrm{Fib}) \rtimes C_2$, using the canonical action exchanging the two factors. The ring $R8_1_22$ is the fusion subcategory of $\mathrm{TY}(C_3) \boxtimes \mathrm{Vec}_{C_4}$ whose simple objects have matching parity: (a, z^{2j}), (m, z) and (m, z^3). Both operations preserve unitarity.

For $R8_1_30$, take the standard unitary $\mathrm{TY}(C_6)$ with bicharacter $\chi(a, b) = \exp(2\pi iab/6)$ and positive parameter $\tau = 1/\sqrt{6}$. Inversion fixes m, preserves every standard associator coefficient, and gives a tensor functor T with identity tensorator. Equip the C_2 action with coherence $T^2 \Rightarrow \mathrm{Id}$ equal to $+1$ on invertibles and -1 on m. This is a T-invariant monoidal natural automorphism, since the TY grading is $\mathbb{Z}/2$. In the equivariantization, the two structures on m have eigenvalues $\pm i$. The tensor decomposition is obtained by splitting the permutation action on each fusion space into its two characters; in particular two odd eigenvalues contribute the additional minus sign. The result is the stated full tensor, with four invertibles, two objects of dimension two, and two of dimension $\sqrt{6}$.

Let $F_{20} = C_5 \rtimes C_4$ with

$$(a, b)(c, d) = (a + 2^b c,\ b + d), \qquad \omega(x, y, z) = (-1)^{b(x)\lfloor (b(y)+b(z))/4 \rfloor},$$

where the two coordinates are reduced modulo five and four, respectively, and representatives $0 \le b < 4$ are used in the cocycle. For $H = \langle (0, 2) \rangle$, the restriction $\omega|_H$ is trivial. The unitary group-theoretical category $\mathcal{C}(F_{20}, \omega, H, 1)$ realizes $R8_1_3$. The supplied code constructs the eight simple homogeneous $\mathbb{C}[H]$-bimodules and their balanced tensor products. All phases are fourth roots of unity and are stored as integer exponents modulo four; the cocycle identity, bimodule coherence, balanced quotient, decomposition multiplicities, and full census tensor match are checked exactly.

The ring $R8_1_6$ has a bosonic quotient construction. Let $\mathcal{D} = PSU(2)_6$, with simples $1, f, \rho, f\rho$, where f is a fermion and $\rho^2 = 1 + \rho + f\rho$. Let $\mathcal{P} = \mathcal{C}(C_4, q)$ be the unitary pointed modular category with $q(z^a) = \exp(\pi i a^2/4)$. Then $b = f \boxtimes z^2$ in $\mathcal{B} = \mathcal{D} \boxtimes \mathcal{P}$ is an order-two boson: its self-braiding is $+1$. It generates a Tannakian $\mathrm{Rep}(C_2)$, whose regular algebra is $A = 1 \oplus b$. The central inclusion supplied by the braiding gives a fusion category $\mathcal{B}_A$ of all right A-modules, with tensor free-module functor F [21, Theorems 8.4.9 and 8.23.3]. The unitary crossed-product construction preserves its positive $*$-structure [39]. The boson is not transparent in $\mathcal{B}$; thus this uses all modules, rather than only local modules, and asserts no descended braiding.

Tensoring by b acts freely on the sixteen simples by $(V_j, z^a) \mapsto (V_{6-j}, z^{a+2})$, with $j \in \{0, 2, 4, 6\}$. By adjunction every free module $F(V_j \boxtimes z^a)$ is simple, and two are isomorphic exactly when their labels belong to the same orbit. Dominance of F gives precisely eight simples. With $g = F(1 \boxtimes z)$ and $r = F(\rho \boxtimes 1)$, their rules are

$$g^4 = 1, \qquad gr = rg, \qquad r^2 = 1 + r + g^2 r.$$

The ordered model basis $(1, g, g^2, g^3, r, gr, g^2 r, g^3 r)$ maps to census labels $(0, 1, 7, 2, 5, 3, 6, 4)$. The verifier reconstructs all coefficients from the truncated $SU(2)_6$ rules and the eight free orbits.

For $R8_1_7$, use either of the two unitary pseudo-$PSU(2)_6$ categories $\mathcal{D}$ of [29, Theorem 4.1]. Their basis $1, g, x, y$ obeys $g^2 = 1$, $gx = y$, $x^2 = y^2 = g + x + y$ and $xy = 1 + x + y$. The two source-bound AnyonWiki associator tables, labelled $(4, 1, 2, 4, 1, 0, 1)$ and $(4, 1, 2, 4, 2, 0, 1)$, are included

in the supplement. Over $\mathbb{Q}(\zeta)$, $\zeta^4+1=0$, the verifier checks all 224 coefficients, 136 invertible associator blocks, unit and rigidity conditions and 2688 pentagons for each table.

Write $a(i)=gi$ and let $F_\ell^{ijk}[e,f]$ take left intermediate e to right intermediate f. In the two tables respectively, the scalars

$$(\lambda_1,\lambda_g,\lambda_x,\lambda_y)=(1,\zeta^2,\zeta,\zeta^3),\qquad (1,-\zeta^2,-\zeta^3,-\zeta)$$

define a half-braiding of g. The exact check is the following identity for each of the 24 fusion channels $N_{ij}^k=1$:

$$\lambda_i\lambda_j F^{igj}_{a(k)}[a(i),a(j)]=\lambda_k F^{gij}_{a(k)}[a(i),k]F^{ijg}_{a(k)}[k,a(j)].$$

Thus $\widetilde{g}=(g,c)$ is an invertible object of $Z(\mathcal{D})$. Its self-braiding satisfies $\lambda_g^2=-1$, so a fermionic condensation argument would be inappropriate. The equivalence between the unitary and ordinary centers permits a unitary representative of this lift [31, corrected Remark 3.1].

Instead take the trivial homomorphism $C_2\to \mathrm{BrPic}(\mathcal{D})$ and modify its product equivalences by the normalized central 2-cocycle $t(1,1)=\widetilde{g}$, with other normalized values the unit. The only nontrivial cocycle identity compares $\widetilde{g}$ with itself, regardless of its order in $\mathrm{Inv}\,Z(\mathcal{D})$. The remaining extension obstruction belongs to $H^4(C_2,\mathbb{C}^\times)=0$ [7, Theorem 1.3]; this is also the trivial-extension zesting of [30, Section 6.2].

This extension is unitary. Choose regular unitary $\mathcal{D}$-bimodule components and unitary multiplication equivalences from the unitary lift. Preliminary associativity intertwiners can be chosen unitary by polar decomposition, so their normalized pentagon defect is $U(1)$-valued. A normalized 3-cochain on C_2 is determined by u at $(1,1,1)$, and its coboundary at $(1,1,1,1)$ is u^2. A unit-modulus square root therefore removes the defect while keeping every coherence map unitary. The resulting regular-component extension is a unitary fusion category; this argument uses no general assertion that arbitrary extensions of unitary categories are unitarizable.

Its simples are (X,a), $a\in C_2$, with

$$(X,a)(Y,b)=(XYg^{ab},a+b).$$

For $z=(1,1)$, this gives $z^2=g$, $zx=xz$ and $x^2=z^2+x+z^2x$. The model basis $(1,g,x,y,z,zg,zx,zy)$ maps to census labels $(0,7,3,4,1,2,6,5)$, and all 512 coefficients agree. The exact associator and central-lift replay is invoked by the finite-update verifier.

The same two rank-four bases have nontrivial flip extensions. For $\mathcal{D}=PSU(2)_6=\mathrm{Ad}(A_7)$, a tensor autoequivalence τ exchanges ρ and $f\rho$ [32, Lemma 4.10]. Its square is tensor-isomorphic to the identity. The fusion rules imply $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{D})=1$, so this square isomorphism is unique after normalization and obeys the C_2 action coherence. Reutter's unitarization and automatic-unitarity theorems give a unitary representative of the entire action [33, Theorems 1–2].

For the second pseudo-$PSU(2)_6$ source table, the permutation $\tau=(x\ y)$ also gives a coherent tensor action, now checked explicitly. Its tensorator J_{ij}^k is 1 when an input is the unit. On the seventeen other admissible channels, ordered lexicographically, write $J=(\sqrt{2})^a\zeta^b$; the exponent vectors are

$$a=(-1,-1,0,-1,-1,-1,-1,-1,0,-1,0,-1,0,-1,0,0,0),$$
$$b=(29,15,10,17,1,5,5,11,0,1,16,15,4,5,0,0,0).$$

All 224 tensor-functor identities hold over $\mathbb{Q}(\zeta)$. The natural isomorphism $\eta:\tau^2\Rightarrow\mathrm{Id}$ has scalars

$$\eta=\left(1,\frac{1+\zeta^2}{2},-\frac{1+\zeta^2}{2},-\frac{1+\zeta^2}{2}\right)$$

in the order $(1,g,x,y)$. The verifier checks all 24 equations $\eta_i\eta_j=\eta_k J_{ij}^k J^{\tau(k)}_{\tau(i),\tau(j)}$ and $\eta_{\tau(i)}=\eta_i$. This is therefore a coherent action, not just a symmetry of the ring. Transport to unitary functors

preserves its coherence; the transported monoidal natural isomorphism is automatically unitary by [33, Theorems 1–2].

For either rank-four base, its nontrivial invertible has a unique central lift: the kernel of $\mathrm{Inv}\, Z(\mathcal{D}) \to \mathrm{Inv}\,\mathcal{D}$ is $\mathrm{Aut}_{\otimes}(\mathrm{Id}_{\mathcal{D}}) = 1$, and a lift was already constructed above. The flip fixes this lift. Thus one may use either the crossed product or its central 2-cocycle modification. The remaining C_2 obstruction is removed by the unitary phase correction described above. With v the nontrivial invertible, their products are

$$(X,a)(Y,b) = \big(X\tau^a(Y)v^{\epsilon ab}, a+b\big), \qquad \epsilon \in \{0,1\}.$$

For $PSU(2)_6$, these give $R8_1_25$ at $\epsilon = 0$ and $R8_1_14$ at $\epsilon = 1$; for pseudo-$PSU(2)_6$, they give $R8_1_42$ and $R8_1_11$, respectively. The exact tensorator and coherent-square replay is `software/verify_pseudo_psu_swap.py`.

The ring $R8_1_23$ is $\mathrm{Vec}_{Q_8}^{C_3}$ for the strict automorphism $i \mapsto j \mapsto k \mapsto i$ of the quaternion group. The fixed objects $1, -1$ each have three equivariant structures, giving six invertibles. The two free orbits $\{i,j,k\}$ and $\{-i,-j,-k\}$ give two simples of dimension three. For a free output orbit, the fusion coefficient is the number of pairs of input group elements with product equal to its chosen representative. For a fixed output, the product fibre is a sum of free C_3-orbits, each contributing its regular character, together with any fixed pairs carrying the product character. These rules independently generate every coefficient. In the order of the three characters over 1, the three over -1, and the two free orbits, the census permutation is $(0,5,6,7,1,2,3,4)$.

The inversion construction for $R8_1_30$ also applies to $A = C_9$ and $A = C_3^2$, producing $R8_1_34$ and $R8_1_35$. Use respectively $\chi(a,b) = \exp(2\pi iab/9)$ and $\chi(a,b) = \exp(2\pi i(a_1b_1 + a_2b_2)/3)$, with $\tau = 1/3$. Inversion preserves these nondegenerate bicharacters, hence the standard unitary TY associators. The parity coherence is again -1 on m and $+1$ on group elements. There are two equivariant simples above 0, four free pairs of nonzero group elements, and two structures with eigenvalues $\pm i$ above m. Their dimensions are $1,1,2,2,2,2,3,3$; the same exact character projection generates the recorded tensors.

Finally take $A = \mathbb{F}_2^2$, $\chi(a,b) = (-1)^{a_1b_2+a_2b_1}$ and $\tau = 1/2$. The whole group $S_3 = \mathrm{GL}_2(\mathbb{F}_2)$ preserves χ, so label permutations give a strict unitary action on the standard TY presentation. For $t = 0, 1$, modify its action coherence on m by

$$\alpha_t(g,h) = (-1)^{tp(g)p(h)},$$

where $p : S_3 \to \mathbb{F}_2$ is parity, leaving the group simples unchanged. This is monoidal by the TY grading, and the action pentagon is the elementary normalized 2-cocycle identity for $p(g)p(h)$. Both equivariantizations are therefore unitary. Their orbits $\{0\}$, $A\backslash\{0\}$ and $\{m\}$ have stabilizers S_3, C_2, S_3. Denote their simples by A_ρ, X_ϵ and M_ρ, where ρ ranges over $1, \mathrm{sign}, \mathrm{std}$ and $\epsilon = \pm$. Above m, the equivariance is $i^{tp(g)}\rho(g)$, so the product of two such factors contributes sign^t.

The fusion coefficients are ordinary character inner products. Multiplication by A_ρ is tensoring with ρ, restricted to C_2 on an X-object. The other rules are

$$\begin{aligned} X_\epsilon X_\delta &= A_{\mathrm{Ind}_{C_2}^{S_3}(\epsilon\delta)} + X_+ + X_-, \\ X_\epsilon M_\rho &= M_{(\mathrm{Ind}_{C_2}^{S_3}\epsilon)\rho}, \\ M_\rho M_\sigma &= A_{\rho\sigma\,\mathrm{sign}^t} + X_{\mathrm{Res}_{C_2}^{S_3}(\rho\sigma\,\mathrm{sign}^t)}, \end{aligned}$$

where subscripts extend additively to reducible representations. For XX, the zero fibre is the induced character and the nonzero fibre is a regular C_2 permutation representation. These formulas give $R8_2_611$ at $t = 0$ and $R8_2_154$ at $t = 1$, both of global dimension 48. The model dimensions in the order $(A_1, A_{\mathrm{sign}}, A_{\mathrm{std}}, X_+, X_-, M_1, M_{\mathrm{sign}}, M_{\mathrm{std}})$ are $(1,1,2,3,3,2,2,4)$. All character arithmetic and tensor comparisons are exact.

One more quadratic model is $R8_1_53$. Yu–Zhang construct a unitary near-group category $NG(G,16)$ for $G=C_4^2$ [34, Proposition 3.1]. Its bicharacter and quadratic refinement may be taken as

$$\chi(g,h)=i^{g_1h_1-g_2h_2},\qquad a(g)=\zeta_8^{3(g_1^2-g_2^2)}.$$

For $H=2G$, exact finite-group arithmetic gives $H=H^\perp$ and $a(h)=\chi(h,g_a)$ with $g_a=(1,1)\notin H$. Izumi's general orbifold theorem applies to $H\subseteq H^\perp$ even when $a|_H$ is nontrivial [35, Theorem 12.5]; its crossed-product-factor construction remains unitary. With $A=G/H=C_2^2$, the resulting rules are

$$\alpha_g\alpha_h=\alpha_{g+h},\qquad \alpha_g\sigma=\sigma\alpha_g,\qquad \sigma^2=\alpha_{g_a}+\sum_{g\in A}\alpha_g\sigma.$$

There are four invertibles and four noninvertibles of dimension $(8+4\sqrt{5})/4=2+\sqrt{5}$, and every noninvertible has a distinct dual. This uses the non-self-dual theorem, rather than its self-dual Lagrangian corollary. In the model order $(00,01,10,11,\sigma,01\sigma,10\sigma,11\sigma)$, the census permutation is $(0,5,6,7,1,3,4,2)$. The portable verifier checks the subgroup hypotheses, the duality shift and all fusion coefficients; the cited theorem supplies categorical existence.

A cyclic version gives $R8_1_12$. Budinski constructs $NG(C_{16},16)$ [36, Theorem 3.1.1 and Table 3.1], with data

$$\chi(x,y)=\exp(2\pi i\,3xy/16),\qquad a(x)=\exp(-\pi i\,3x^2/16).$$

Although the table prints decimal arguments, the thesis's Section 3.3.2 gives an exact algebraic certification, with the defining-polynomial scope explicitly including $n=16$. We use that exact existence statement as a citation dependency; no numerical reconstruction is counted here as a category. For $H=4C_{16}$, one has $H=H^\perp$ and $a(h)=\chi(h,2)$. The same unitary orbifold theorem [35, Theorem 12.5] therefore gives

$$g^4=1,\qquad \sigma g=g^{-1}\sigma,\qquad \sigma^2=g^2+(1+g+g^2+g^3)\sigma.$$

In the order $(1,g,g^2,g^3,\sigma,g\sigma,g^2\sigma,g^3\sigma)$, the census permutation is $(0,1,7,2,3,6,4,5)$. The four noninvertibles are paired by duality, distinguishing this ring from the standard C_4 Haagerup–Izumi ring. The finite verifier checks all subgroup hypotheses and the full tensor exactly.

The ring $R8_1_83$ has a triality-gauging model. The S_3 gauging of the three-fermion category exists by [2, Section 6.2]; the unobstructed gauging data form the usual $H^3(S_3,U(1))$ torsor, including the twist denoted $m=0$ in [37, Section VI.D and Table IX]. This supplies an actual category. Starting from a pointed category, gauging produces a weakly group-theoretical fusion category, hence one admitting a unitary realization [38, Theorem 5.20]. Its eight integral simples, with dimensions $(1,1,2,3,3,4,4,4)$, form a fusion subcategory of dimension 72.

The integer fusion formulas for this subcategory are independently checked against the full twelve-object S-matrix of [37, Eq. (6.53), $m=0$]. Its entries lie in $\mathbb{Q}(\sqrt{2})$; using the unnormalized matrix, the verifier checks $S^2=144I$, all 1728 nonnegative integral Verlinde coefficients, and fusion closure of the eight integral simples. It then compares the entire tensor with the pinned census under the stored permutation. Category existence comes from the gauging construction, not merely from the feasible modular datum.

The table provides one construction per positive tensor. Some have two isomorphic presentations; `data/realized_models.json` records every matched constructor, exact model tensor, precise model source and source-to-model isomorphism. Labels such as "pseudo-$PSU(2)_6$" denote the conventional fusion-ring name, not a weaker existence assertion: the cited rank-four table supplies a unitary category [14]. No undecided ring is in this table.

4.2. **Pointed equivariantizations and an affine group.** Four further models are strict unitary equivariantizations of untwisted pointed categories. Their group automorphisms are

$$\begin{aligned}
&R8_2_244: && A=C_4^2, && (x,y)\mapsto(-y,x-y), && G=C_3;\\
&R8_2_289: && A=C_2^4, && (x,y,z,w)\mapsto(y,x+y,w,z+w), && G=C_3;\\
&R8_2_143: && A=C_{17}, && x\mapsto 4x, && G=C_4;\\
&R8_2_179: && A=C_2^4, && (x,y,z,w)\mapsto(w,x+w,y+w,z+w), && G=C_5.
\end{aligned}$$

In each case the displayed generator has the stated order and acts freely off zero. The zero orbit supplies the ordinary characters of G, and each nonzero orbit has trivial stabilizer. Tensor products are calculated by orbit addition and projection to stabilizer characters, giving the exact full source tensors. Strict permutation functors with identity tensorators supply coherent unitary actions.

The ring $R8_2_237$ is $K_0(\mathrm{Rep}(A\Gamma L(1,8)))$. This group of order 168 has elements (v,a,t) acting on $\mathbb{F}_8$ as $x\mapsto v+ax^{2^t}$. Its simple dimensions are $(1^3,3^2,7^3)$. The three dimension-seven characters have the exact expression

$$\chi_s(v,a,t)=\zeta_3^{st}\sum_{\substack{r\in\mathbb{F}_8^\times\\ ar^{2^t-1}=1}}(-1)^{\mathrm{Tr}_{\mathbb{F}_8/\mathbb{F}_2}(v/r)}.$$

Together with the characters inflated from $\mathbb{F}_8^\times\rtimes C_3$, they give an exact character table in $\mathbb{Q}(\zeta_{21})$. Orthogonality and all tensor multiplicities are checked independently by finite group sums.

4.3. **Bosonic quotients and the D_{14} model.** The ring $R8_2_557$ is the central bosonic quotient of $PSU(2)_6\boxtimes PSU(2)_6$ by its diagonal invertible. Each factor invertible is a fermion, so their product has self-braiding $+1$ and trivial pointed associator. It generates a genuine central $\mathrm{Rep}(C_2)$. All sixteen simples have free diagonal orbits; the eight free-module orbits give the stated ring.

For $R8_2_313$, use instead the two unitary pseudo-$PSU(2)_6$ categories of [29], with the exact sourced associators discussed above. Their central invertibles have self-braidings $+i$ and $-i$, and each has associator $F_{ggg}=-1$. The diagonal product therefore has both trivial self-braiding and trivial associator; its central $\mathrm{Rep}(C_2)$ gives the same eight-orbit quotient construction. For each rank-four factor the universal grading is trivial, so a central lift of an invertible is unique; its central square is the unit. This verifies the order-two condition as well as the associator and self-braiding conditions required for the categorical quotient.

The ring $R8_2_855$ is the even D_{14} category, or the local module category of the $SU(2)_{24}$ algebra $1\oplus V_{24}$. It is unitary modular [47, 48]. Its basis is $f_0,f_2,f_4,f_6,f_8,f_{10},P,Q$, with P,Q self-dual. Fold the bulk $SU(2)_{24}$ products by $f_{2t}=f_{2(12-t)}$ and $f_{12}=P+Q$. The remaining rules are

$$Pf_{2j}=\sum_{k=6-j}^{5}f_{2k}+\begin{cases}P & j\text{ even},\\ Q & j\text{ odd},\end{cases}\qquad Qf_{2j}\text{ interchanges }P,Q,$$

$$P^2=P+f_0+f_4+f_8,\quad Q^2=Q+f_0+f_4+f_8,\quad PQ=f_2+f_6+f_{10}.$$

All these constructions preserve the unitary structure.

4.4. **A Morita construction over C_7.** The ring $R8_2_239$ is a Morita dual of an S_3 extension of Vec_{C_7}. In $\mathrm{BrPic}(\mathrm{Vec}_{C_7})=O(C_7\oplus\widehat{C_7})$ take $r=\mathrm{diag}(2,4)$ and let s exchange the two coordinates. They generate S_3. Its degree-three obstruction vanishes by coprime cohomology with seven-primary coefficients; the next obstruction lies in $H^4(S_3,\mathbb{C}^\times)=0$. Thus the actual extension exists [7, Theorem 1.3]. Its C_3 restriction is pointed, on $F_{21}=C_7\rtimes C_3$. The associator class there is inflated from C_3: the order-three action kills the $H^3(C_7,\mathbb{C}^\times)$ invariant part. Surjectivity of $H^3(S_3,\mathbb{C}^\times)\to H^3(C_3,\mathbb{C}^\times)$ permits an outer twist that makes this restriction untwisted.

Take the separable algebra $B=\bigoplus_{h\in C_3}h$ in this restriction. The category of B-bimodules is an actual Morita dual, whose neutral category is $\mathrm{Rep}(F_{21})$. Write its simples as α_i $(i\in\mathbb{Z}/3)$, X,Y,

and odd m_j ($j \in \mathbb{Z}/3$). The nontrivial rules, with all m_j self-dual, are

$$X^2 = X + 2Y, \quad Y^2 = 2X + Y, \quad XY = \sum_i \alpha_i + X + Y,$$

$$\alpha_i m_j = m_{i+j}, \quad m_j \alpha_i = m_{j-i}, \qquad X m_j = Y m_j = m_j X = m_j Y = \sum_k m_k,$$

$$m_j m_k = \alpha_{j-k} + X + Y.$$

They follow from the S_3 bimodule action and reciprocity, and match all 512 source coefficients. The extension and its Morita dual are weakly group-theoretical, hence unitarizable [38, Theorem 5.20].

4.5. **Charge-conjugation defects of $PSU(3)_4$.** The ring $R8_2_290$ is a unitary C_2 crossed extension of the rank-five modular category $\mathcal{B} = PSU(3)_4$. Charge conjugation is an actual braided symmetry [49, Corollary 2.7]. The invertible group of $\mathcal{B}$ is trivial, so the degree-three gauging obstruction vanishes, and $H^4(C_2, U(1)) = 0$ removes the remaining obstruction. The unitary gauging theorem [2, Sections 3.2–3.4] therefore supplies an actual unitary extension. Its odd rank is three, the number of charge-conjugation-fixed neutral simples [50, Proposition 1.1].

For completeness, its tensor is forced by a small exact module calculation. In the neutral basis $(0,0), (3,0), (1,1), (0,3), (2,2)$ the nonselfdual generator X has regular characteristic polynomial $(t^2 + t + 2)(t^3 - 2t^2 - t + 1)$, computed by quantum Weyl alternants in $\mathbb{Q}(\zeta_{21})$. Its odd action A has the cubic factor as characteristic polynomial, since its Perron root has degree three. The commuting neutral action and the dual-transpose relation make A normal. Its three roots are real, so A is symmetric. Nonnegative integer entries, $\mathrm{Tr}(A) = 2$ and $\mathrm{Tr}(A^2) = 6$ give, up to relabelling,

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

The five neutral actions are $I, A, A^2 - I, A, A^2 - A$. Distinct odd dimensions force each odd object to be self-dual; reciprocity then determines every odd product. The independent constructor matches all 512 census coefficients.

4.6. **A self-dual Hopf extension in dimension thirty-two.** For $R8_2_575$ the neutral Hopf algebra is $H = H_{C:\sigma_1}$ of dimension sixteen from [43]. A symmetric nondegenerate Hopf pairing supplies the extra data of the Davydov–Runkel construction [51, Theorem 2.5]. Here is enough explicit information to fix the construction and its normalization. Write $h(p,q,j)$ for its basis, with p modulo four and q, j modulo two. Put $\zeta = \exp(2\pi i/16)$ and $i = \zeta^4$. The pairing is

$$P(h(p,q,j), h(r,s,k)) = \delta_{s,j}\delta_{k,q}\, i^{-3rp}\zeta^{jq}/4.$$

The supplementary Hopf presentation and checker verify nondegeneracy, symmetry and every pairing identity exactly in $\mathbb{Q}(\zeta)$. Davydov–Runkel use $H^\vee = (H^*)^{\mathrm{op,cop}}$; consequently their self-duality map is $\Gamma = S \circ P^{-1}$, rather than the unmodified inverse pairing. The right cointegral $\lambda(h(p,q,j)) = \delta_{j,0}/2$ satisfies $\Gamma(\lambda) = 4\Lambda$ and $\lambda(\Gamma(\lambda)) = 1$, where Λ is the normalized integral. The distinguished grouplike is one, $S^2 = 1$, and Γ is symmetric, so all conditions of the cited theorem hold. Its rigidity criterion [51, Proposition 3.25] gives an actual fusion category $\mathrm{Rep}(H) \oplus \mathrm{Vec}$. Its neutral ring is $\mathrm{Rep}(D_{16})$; for the odd object M of dimension four,

$$XM = MX = \mathrm{FPdim}(X)M, \qquad M^2 = \sum_{X \in \mathrm{Irr}(H)} \mathrm{FPdim}(X)X.$$

The dimension-32 solvability theorem and [38, Theorem 5.20] make this fusion category unitarizable. The complete model tensor, Hopf identities, and source comparison are separate exact checks.

4.7. **The dual Haagerup–Izumi C_7 category.** The ring $R8_2_868$ is the dual even part of the unitary Haagerup–Izumi C_7 subfactor. Its existence and center are supplied by [52, Theorem 5(d)]. Put $d = (7+\sqrt{53})/2$, so $d^2 = 1+7d$. Izumi's dual Q-system analysis [8, Proposition 7.4 and Theorem 7.5] gives rank eight, with dimensions $1, d, (d-1)^3, (d+1)^3$.

We verify the canonical-algebra hypothesis needed to use the full fusion rules in [52, Theorem 7]; matching modular data alone would not suffice. In its published double, the twist-one simples are 1, b of dimension d^2, and six A_i of dimension $1+d^2$. The nonzero C_{53} sector has no twist one, and the other twist-one labels are precisely those with zero second coordinate. For the actual dual category $\mathcal{D}$, its canonical Lagrangian algebra $L = I_{\mathcal{D}}(1)$ is connected, has trivial twist and dimension $7(1+d^2)$. Therefore

$$L = 1 + m_b b + \sum_{i=1}^{6} m_i A_i.$$

Writing $T = \sum_i m_i$, rational and irrational coefficients in the dimension identity give $m_b + T = 7$ and $m_b + 2T = 13$, hence $m_b = 1, T = 6$. Adjunction gives $\dim \mathrm{End}(L) = \mathrm{rank}(\mathcal{D}) = 8$, so $\sum_i m_i^2 = 6$. Thus every $m_i = 1$. The actual canonical algebra has exactly the underlying object required by the cited theorem; no assumption about a guessed algebra multiplication is made.

For a direct exact constructor, use basis $1, a, b_1, b_2, b_3, c_1, c_2, c_3$. Put $B = \sum_i b_i$, $C = \sum_i c_i$, $b_0 = c_0 = 0$, and $b_{-i} = b_i, c_{-i} = c_i$ for indices modulo seven. The rules are

$$\begin{aligned}
a^2 &= 1 + a + B + C, & ab_i &= a + B - b_i + C,\\
ac_i &= a + B + c_i + C, & b_i c_j &= a + B + C,\\
b_i b_j &= \delta_{ij} 1 + (1-\delta_{ij}) a + B - b_{i+j} - b_{i-j} + C,\\
c_i c_j &= \delta_{ij} 1 + (1+\delta_{ij}) a + B + c_{i+j} + c_{i-j} + C.
\end{aligned}$$

They match every source coefficient; all 4096 associativity equations and all dimension products in $\mathbb{Q}(d)$ are checked. Both even parts of the cited finite-depth C^* subfactor are unitary.

Model	Model-to-census permutation
$R8_2_143$	$(0,1,7,2,3,5,6,4)$
$R8_2_179$	$(0,1,4,3,2,5,6,7)$
$R8_2_237$	$(0,1,2,5,6,7,4,3)$
$R8_2_239$	$(0,1,2,3,4,5,7,6)$
$R8_2_244$	$(0,1,2,3,7,4,5,6)$
$R8_2_289$	$(0,1,2,3,4,5,6,7)$
$R8_2_290$	$(0,1,6,2,7,3,4,5)$
$R8_2_313$	$(0,7,1,2,3,4,6,5)$
$R8_2_557$	$(0,1,2,7,3,6,5,4)$
$R8_2_575$	$(0,5,1,2,3,7,4,6)$
$R8_2_855$	$(0,1,7,4,5,6,2,3)$
$R8_2_868$	$(0,3,1,2,4,5,7,6)$

The model bases are fixed by the independent constructors in the supplement. Each permutation is checked on the entire ordered tensor; it is not inferred from a character table or dimension vector.

5. Scoped obstructions inside the general frontier

The following statements refine the frontier without changing any general exclusion count. The exact source subrings, grading supports and fixed objects are checked separately from the general-decision certificates.

Proposition 5.1. *The rings R8_2_24, R8_2_152, R8_2_153, R8_2_166, R8_2_495, R8_2_315, R8_2_317 have no unitary categorification. The ring R8_2_312 has no pseudo-unitary categorification. They remain generally unresolved.*

Proof. The first five rings contain the rank-four $2D2$ even ring. Its unitary categorification is unique [29, Theorem 3.43]. Every invertible bimodule over this category is an automorphism bimodule [54, Theorem 4.19], hence is regular as a left module. For $R8_2_152$, $R8_2_153$, $R8_2_166$, $R8_2_495$, the odd component of a faithful C_2 grading has no invertible simple. A regular module component has the regular dimension ratios, and its equal total FP dimension fixes the scaling to one; it must contain an invertible. This is a contradiction.

For $R8_2_24$, the odd component contains an invertible u of order four with $u^2 = g$, where g is the nontrivial neutral invertible. Commutativity of the tensor makes $\mathrm{Ad}(u)$ fix every neutral label, while $\mathrm{Ad}(u)^2 = \mathrm{Ad}(g)$. The label-fixing kernel of the neutral tensor-automorphism group is C_2^2: Grossman's computation gives an order-eight automorphism group and a nontrivial $\mathrm{Ad}(g)$, and the outer, label-fixing order-two action of [29, Corollary 3.45] commutes with it. These two independent involutions exhaust the order-four kernel. Thus no label-fixing automorphism squares to $\mathrm{Ad}(g)$.

The rings $R8_2_315$, $R8_2_317$ contain ordinary and pseudo-$PSU(2)_6$ subcategories with the same nontrivial invertible g. In the unitary classifications their second indicators are respectively $+1$ and -1. The canonical spherical structure restricts to the canonical one on each unitary fusion subcategory, so the indicator of the shared object cannot disagree.

Finally $R8_2_312$ contains a pseudo-$PSU(2)_6$ subring and an additional g-fixed simple. Larson's pseudo-unitary analysis [55, Proposition 5.5 and proof of Theorem 5.8, Case 1] forces $\nu_2(g) = -1$, hence the nontrivial associator on its pointed C_2 subcategory. The g-fixed simple gives a rank-one module over that pointed category, forcing the associator class to be trivial by its scalar module pentagon. This proves the pseudo-unitary contradiction. None of the preceding arguments supplies a general exclusion for these eight rings. □

6. Reproduction and mathematical limits

For the frozen release, run `python3 tools/verify_upper_residual_20261002b.py` from the repository root, and repeat with `-O`. The archive dispatcher supplies the same source-bound stage. It reconstructs the new model tensors, verifies the new general and scoped certificates, checks their source identities, and verifies preservation of the earlier ledger decisions. All checks use exact integer or algebraic arithmetic; no numerical fit supplies a categorification decision.

The historical full-certificate runs, including the exhaustive signed proofs and appendix tensor transfers, retain their original reports. Historical constructor updates are automatically replayed against isolated copies of their original ledgers and model catalogues. Those entry points are not discovery commands for the expanded current catalogue. Preservation evidence does not claim a fresh execution of every earlier costly search. Original census analysis programs are retained for provenance and are not needed to verify the frozen release. A timeout or an unfinished branch never constitutes an exclusion.

The three generally open multiplicity-one rings $R8_1_9$, $R8_1_18$, $R8_1_49$ each have two FP dimensions, 1 and $2 + \sqrt{5}$. Their noninvertibles form one orbit under the invertible action. The published formulation of [24, Lemma 3.1] says that any categorification with two FP dimensions is Galois conjugate to a pseudo-unitary one. Consequently, for these rings, existence as a general, pivotal, spherical or pseudo-unitary category is equivalent. This does not say that every categorification is pseudo-unitary, and gives no unitary existence claim. The irrational braided generalized-near-group classification [53, Theorem 3.6] rules out a braiding on any of these three rings. Their general categorification remains unresolved. The known order-four Haagerup–Izumi Morita families [54, Theorem 4.13 and Lemma 4.7] give no matching model; this exhaustion concerns those families, not all fusion categories.

This verification establishes the finite applications *conditional on* the cited theorems and the source census completeness. It does not settle the remaining 55 generally undecided rings, certify a multiplicity-three census, or classify categories up to tensor equivalence. Some criteria first exclude rings at multiplicity two, but nothing here implies that no new criterion is needed beyond that bound. Noncommutative Extended Haagerup models at rank eight have multiplicities four and seven [42]; they are genuine models but outside the present complete census bound.

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